Intrusion Detection System (IDS)

LetsDefend SOC Analyst lesson | Asha Latha Panuganti | October 2026

Summary

Hands-on lab analysing Zeek IDS logs. I read the HTTP and FTP logs to identify who connected to whom, over which protocol, and what action was performed. An IDS detects and alerts but does not block, so these logs are the starting point for an analyst's triage.

Objectives

- Understand IDS types (NIDS, HIDS, PIDS, APIDS, hybrid), their placement and their role: detect and alert, not block.
- Parse structured Zeek JSON logs to identify the HTTP request method and the FTP command used.

Environment and tools

LetsDefend lab VM (Windows), Notepad++, Zeek `http.log` and `ftp.log` (JSON format).

Method

- Extracted the zipped logs and opened them in Notepad++.
- Read each field by name (`id.orig_h`, `id.resp_p`, `method`, `command`) instead of guessing by position.
- Used port numbers to decide which side of a connection was the server and which protocol was involved.
- Cross-checked each finding against the status and reply codes to confirm the outcome.

Findings

Log	Key fields observed	Finding
Zeek HTTP	192.168.33.87:46378 to port 80; host letsdefend.io; uri /; status 200	HTTP **GET** request answered with 200 OK (39-byte text/plain body).
Zeek FTP	192.168.74.100 to port 21; command RETR; reply 226 "Transfer complete"; file_size 74	File downloaded with the FTP **RETR** command; transfer completed.

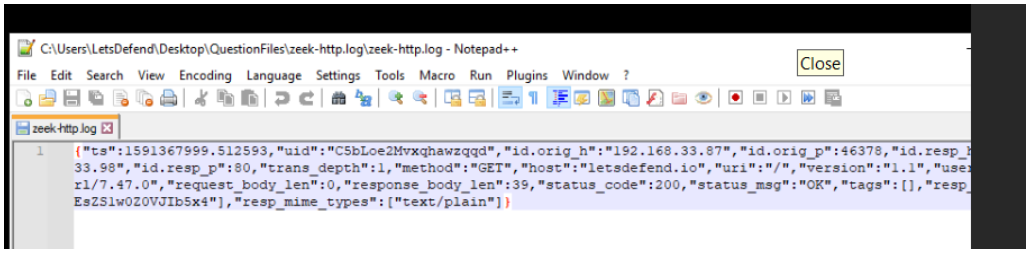


Figure 1: Zeek HTTP log (method field: GET).

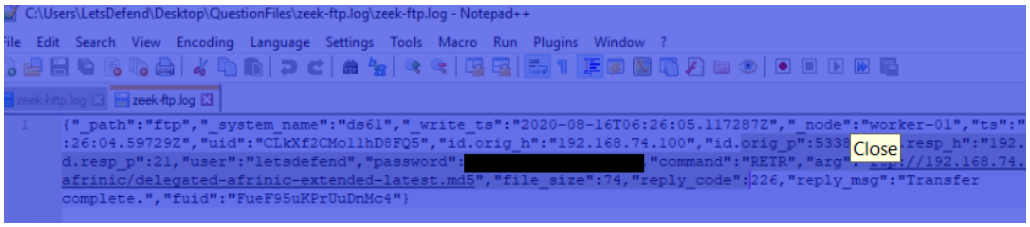


Figure 2: Zeek FTP log (command RETR, reply code 226). Credentials redacted.

Security observations

- **FTP exposes credentials.** The Zeek FTP log recorded the username and password in cleartext, so FTP should be replaced with SFTP or FTPS.
- **An IDS only detects.** Detection quality depends on rule quality, so rules need tuning to reduce false positives and missed attacks, and alerts should feed a SIEM.
- **Logs need context.** A GET with a 200 or a successful RETR is normal on its own. It becomes interesting only with the host, the file requested and the traffic source.

Skills demonstrated

Log analysis, network protocols (OSI model, HTTP, FTP), IDS concepts, reading JSON logs, documenting findings.

Next steps

- Write a custom Snort or Suricata rule that alerts on cleartext FTP logins and test it against a PCAP.
- Forward the Zeek logs into Wazuh and chart HTTP methods and FTP commands.